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A Capitalized Title: Something about a Package foo

FirstName LastName
University/Company

Second Author
Affiliation

Third Author
Universitat Autònoma
de Barcelona

Abstract

Blah, blah, blah... We introduce **funres**, an R and Python package for constructing and visualizing functional residuals for various types of statistical models

Keywords: keywords, not capitalized, Java.

1. Introduction

This template demonstrates some of the basic LaTeX that you need to know to create a JSS article.

1.1. Code formatting

In general, don't use Markdown, but use the more precise LaTeX commands instead:

- Java
- **plyr**

One exception is inline code, which can be written inside a pair of backticks (i.e., using the Markdown syntax).

If you want to use LaTeX commands in headers, you need to provide a `short-title` attribute. You can also provide a custom identifier if necessary. See the header of Section 2 for example.

2. R code

Can be inserted in regular R markdown blocks.

```
R> x <- 1:10
R> x

 [1]  1  2  3  4  5  6  7  8  9 10
```

2.1. Features specific to rticles

- Adding short titles to section headers is a feature specific to **rticles** (implemented via a Pandoc Lua filter). This feature is currently not supported by Pandoc and we will update this template if it is officially supported in the future.
- Using the `\AND` syntax in the `author` field to add authors on a new line. This is a specific to the `rticles::jss_article` format.

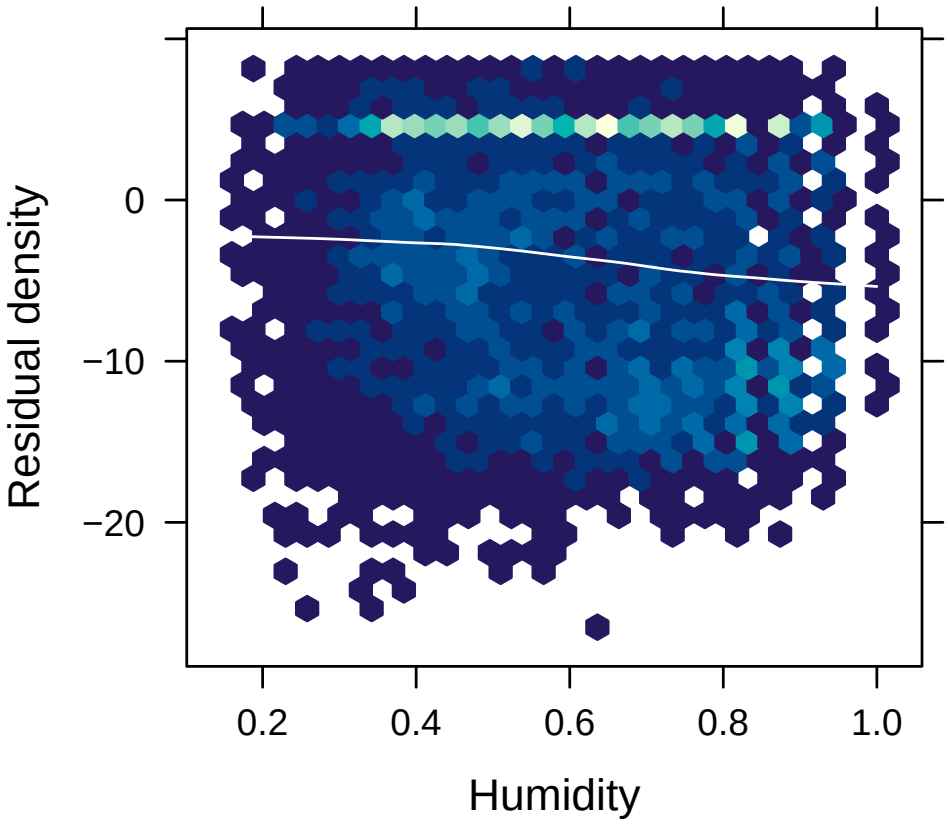
2.2. Bike sharing

Blah, blah, blah…

```
R> library(funres)
R>
R> # Load bike sharing data set
R> bike <- read.csv("data/bike_hour.csv")
R> bike <- bike[bike$yr == 1, ]  # year one
R> bike$winter <- as.factor(ifelse(bike$season == 1, "yes", "no"))
R>
R> # Baseline GLM model
R> bike.glm <- glm(cnt ~ winter + hr + workingday + weathersit + temp +
+                    hum + windspeed, family = "poisson",data = bike)
R> summary(bike.glm)


Call:
glm(formula = cnt ~ winter + hr + workingday + weathersit + temp +
    hum + windspeed, family = "poisson", data = bike)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-26.468  -10.306   -3.314    4.498   39.443

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.8480890  0.0048631  996.91   <2e-16 ***
```

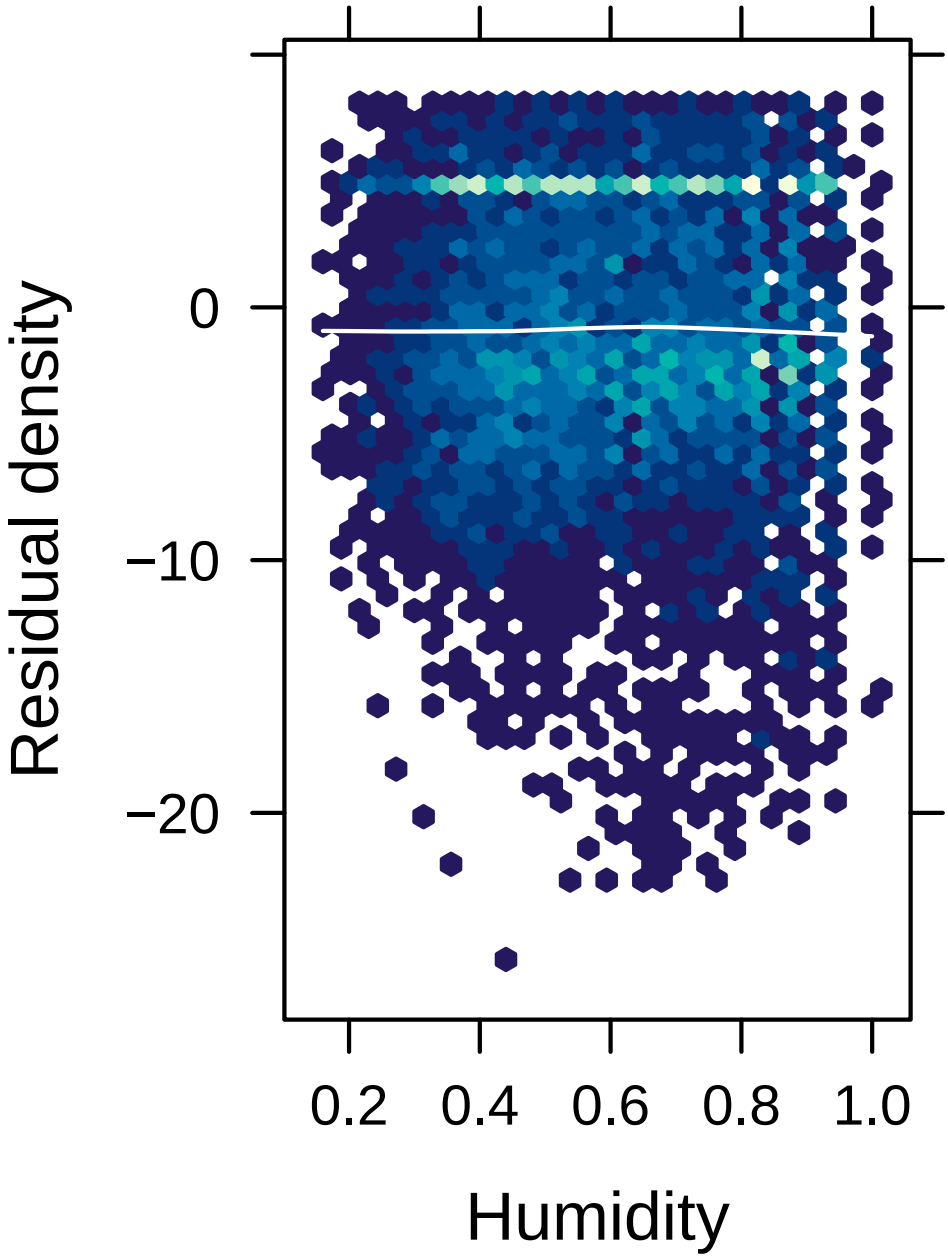
```
winteryes   -0.3230338  0.0022707 -142.26   <2e-16 ***
hr           0.0454776  0.0001149  395.63   <2e-16 ***
workingday   0.0608633  0.0015271   39.86   <2e-16 ***
weathersit  -0.0253463  0.0013583  -18.66   <2e-16 ***
temp         1.1571469  0.0045315  255.35   <2e-16 ***
hum         -1.0045842  0.0045746 -219.60   <2e-16 ***
windspeed    0.2229130  0.0060893   36.61   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1647291  on 8733  degrees of freedom
Residual deviance: 1070670  on 8726  degrees of freedom
AIC: 1128549

Number of Fisher Scoring iterations: 5
```

```
R> fredplot(bike.glm, x = bike$hum, xlab = "Humidity", scale = "normal", type = "hex")
```




```
R> # xnames <- c("winter", "hr", "temp", "hum")
R> # plots <- lapply(xnames, FUN = function(xname) {
R> #   x <- bike[[xname]]
```

```
R> #   if (is.factor(x)) x <- jitter(as.integer(x))
R> #   fredplot(bike.glm, x = x, xlab = xname, scale = "normal")
R> # })
R> # gridExtra::grid.arrange(grobs = plots, ncol = 2)

R> library(mgcv)

Loading required package: nlme

This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.

R> # GAM model
R> bike.gam <- gam(cnt ~ winter + s(hr) + workingday + weathersit + s(temp) +
+                    s(hum) + s(windspeed), family = "poisson",data = bike)
R> summary(bike.gam)


Family: poisson
Link function: log

Formula:
cnt ~ winter + s(hr) + workingday + weathersit + s(temp) + s(hum) +
    s(windspeed)

Parametric coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  5.200771   0.002614 1989.82   <2e-16 ***
winteryes   -0.282128   0.002359 -119.57   <2e-16 ***
workingday   0.055380   0.001538   36.02   <2e-16 ***
weathersit  -0.138515   0.001447  -95.72   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
               edf Ref.df Chi.sq p-value
s(hr)        8.995  9.000 523875  <2e-16 ***
s(temp)      8.474  8.888  46540  <2e-16 ***
s(hum)       8.983  9.000   6572  <2e-16 ***
s(windspeed) 8.940  8.998   1428  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.721   Deviance explained = 77.4%
UBRE = 41.603  Scale est. = 1         n = 8734

R> fredplot(bike.gam, x = bike$hum, xlab = "Humidity", scale = "normal", type = "hex")
```


Residual density
0
−10
−20
0.2
0.4
0.6
0.8
1.0
Humidity

Affiliation:

FirstName LastName
University/Company
First line
Second line
E-mail: name@company.com
URL: https://posit.co

Third Author
Universitat Autònoma de Barcelona
Department of Statistics and Mathematics,
Faculty of Biosciences,
Universitat Autònoma de Barcelona

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